Supplementary Materials

Manuscript: Spliceosome activation defines an epigenetically hyperactive and therapeutically targetable malignant cell state in glioblastoma

This document describes all supplementary figures and tables.

Supplementary Figures

Figure S1 | CopyKAT purity control. (A) Aneuploid (tumor) fraction per sample estimated by copy-number inference (18 samples, 16,255 cells called). (B) WEBS distribution across LLS/LDTS/LHS within aneuploid-only cells (n = 4,421; Kruskal?Wallis p < 10?¹?), demonstrating that WEBS stratification is not an artifact of non-malignant admixture.

Figure S2 | Parameter sensitivity. (a) WEBS computed at UCell maxRank from 5% to 90% of genes vs the default (1,500 genes); WEBS is stable up to maxRank ? 0.10 and drifts at larger values, motivating the fixed maxRank = 1,500 setting. (b) Leave-one-out and leave-two-out removal of writer/eraser genes (57 perturbations; median Spearman ? = 0.95 vs default). (c) Cross-platform consistency of patient-level WEBS between Smart-seq2 and 10x in GSE131928 (? = 0.67, n = 10 shared patients).

Figure S3 | Method benchmarking and computational performance. Patient-level WEBS vs AUCell (? = 0.95), ssGSEA (? = 0.87), all-gene mean (? = 0.92); PLAGE is sign-inverted (? = ?0.97, a known PCA orientation artifact). Runtime and memory scaling at 1k?35.8k cells: WEBS completes 35,823 cells in 213 s with <2 MB incremental memory; ssGSEA exceeds available memory beyond ~25k cells.

Figure S4 | Pan-cancer scope. WEBS?survival associations across TCGA cohorts (GBM, LUAD, BRCA, SKCM): significant in LUAD (HR = 0.64, p = 0.039) but not in BRCA/SKCM/GBM-bulk, delineating the cancer-type-specific scope of the axis.

Figure S5 | Bootstrap stability of the LASSO-Cox feature selection. Selection frequency of candidate genes across 100 bootstrap resamples of the training cohort; HDAC1 (100%), LDHB (97%), VEGFA (91%), LDHA (89%) are the most stable features.

Figure S6 | SHAP interpretability of the six-gene model. Mean |SHAP| per gene overall and by risk group; HDAC1 is the dominant contributor (protective direction, consistent with its eraser role).

Figure S7 | Spliceosome specificity control. Layer discordance (protein ? ? mRNA ? vs WEBS) for the spliceosome (0.82) vs 100 size-matched random gene sets (empirical p = 0.010) and the proteasome as a secondary housekeeping-complex case (0.77).

Figure S8 | Spliceosome acetylation-site heatmap. Top 40 differentially acetylated spliceosome sites (limma, FDR < 0.05), samples ordered by WEBS group; site-level acetylation is enriched in the writer-dominant state (OR = 7.36, p = 1.3 × 10?¹?).

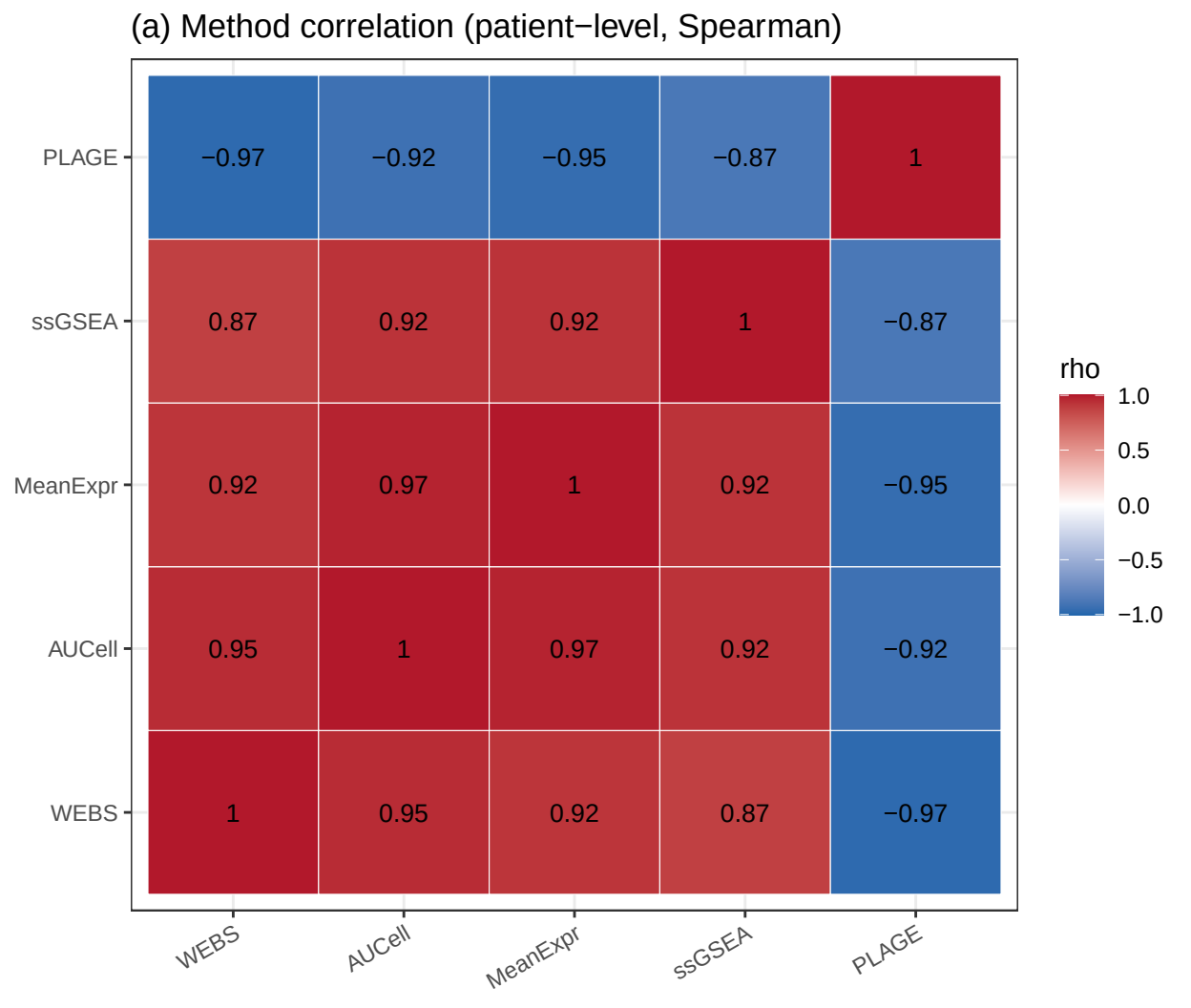
Supplementary Tables (Supplementary_Tables.xlsx)

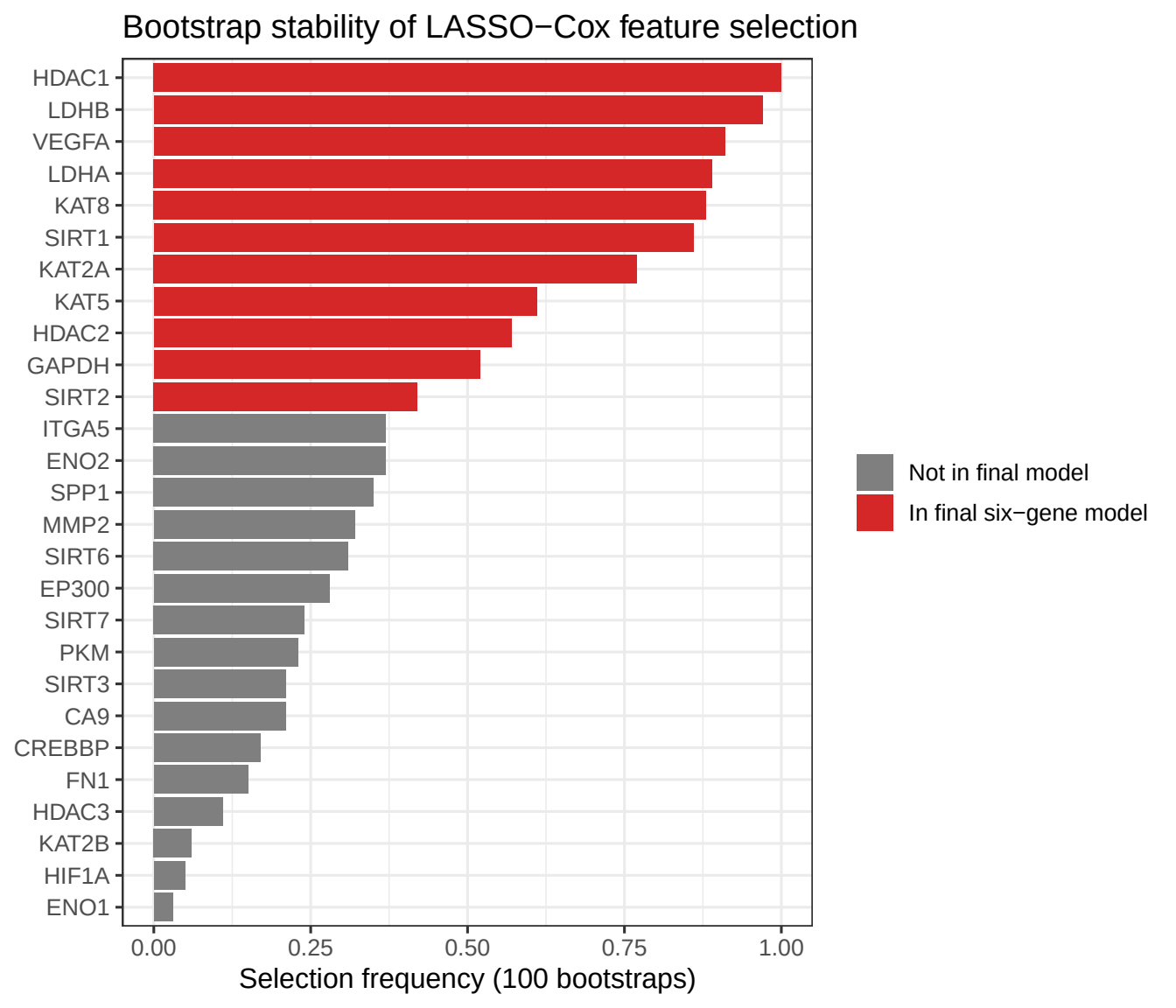
Sheet	Content
S_CPTAC_WEBS	Protein-level WEBS and group assignment for 99 CPTAC-GBM samples

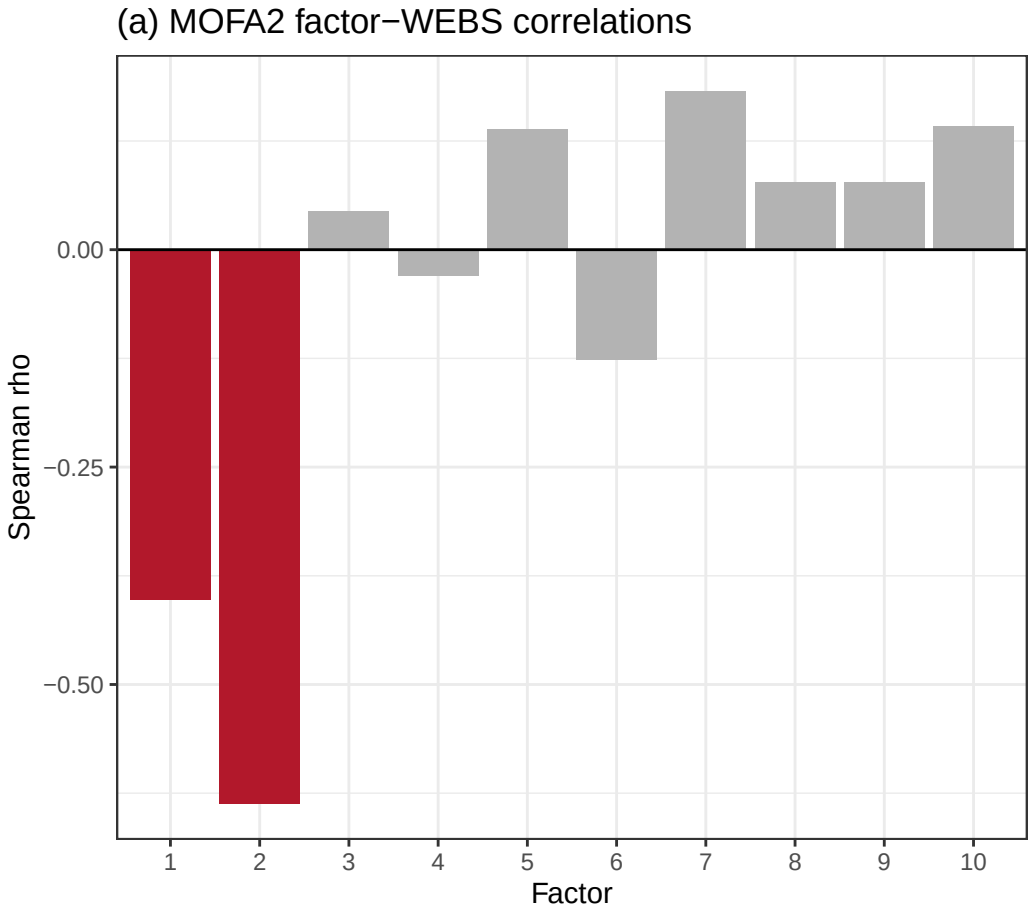
S_CPTAC_WEBS	Protein-level WEBS and group assignment for 99 CPTAC-GBM samples
S_CPTAC_limma	Full limma differential abundance results (11,141 proteins, LHS vs LLS)
S_spliceosome	Spliceosome protein subset results (111 quantified, 107 up-regulated)
S_acetyl_sites	Site-level acetylome differential analysis (8,350 sites)
S_acetyl_corr	Per-site acetylation?WEBS correlations
S_bulk_survival	WEBS survival analysis across bulk cohorts
S_signature_valid	Six-gene signature validation across cohorts (fixed coefficients)
S_bootstrap	Bootstrap selection frequencies (100 resamples)
S_SHAP	SHAP values summary per gene and risk group
S_pancancer	Pan-cancer WEBS?survival summary (4 TCGA cohorts)
S_benchmark	Method correlation matrix (WEBS/AUCell/ssGSEA/PLAGE/mean)
S_performance	Computational runtime and memory at 1k?35.8k cells
S_sensitivity_maxrank	UCell maxRank sweep
S_sensitivity_loo	Leave-one/two-out robustness (57 perturbations)
S_specificity	Spliceosome vs random-pathway discordance control
S_spatial	Moran's I / Geary's c spatial autocorrelation per slice
S_MOFA2	MOFA2 factor?WEBS correlations (4-view CPTAC integration)
S_DepMap	WEBS?dependency correlations for candidate targets
S_GDSC2_drugs	WEBS?sensitivity correlations across 286 drugs
S_GDSC2_HDAC	HDAC-inhibitor subset results
S_combinations	Bliss-independence combination predictions
S_CMap	Drugs targeting LHS-up proteins with WEBS-linked sensitivity
S_CellChat	Pathway-level information flow LLS vs LHS
S_MIF_perm	MIF?CD74/CXCR4 permutation test
S_docking	Molecular docking binding energies (4 target?ligand pairs)

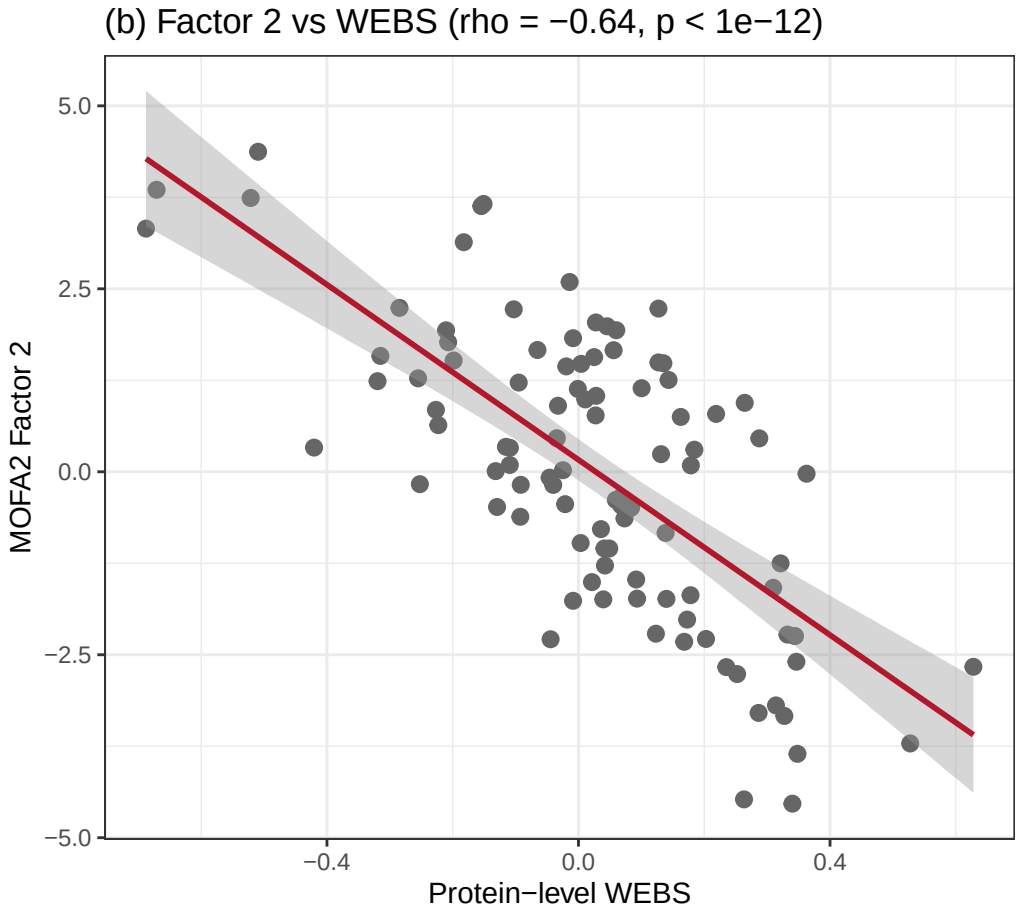
Supplementary Methods notes

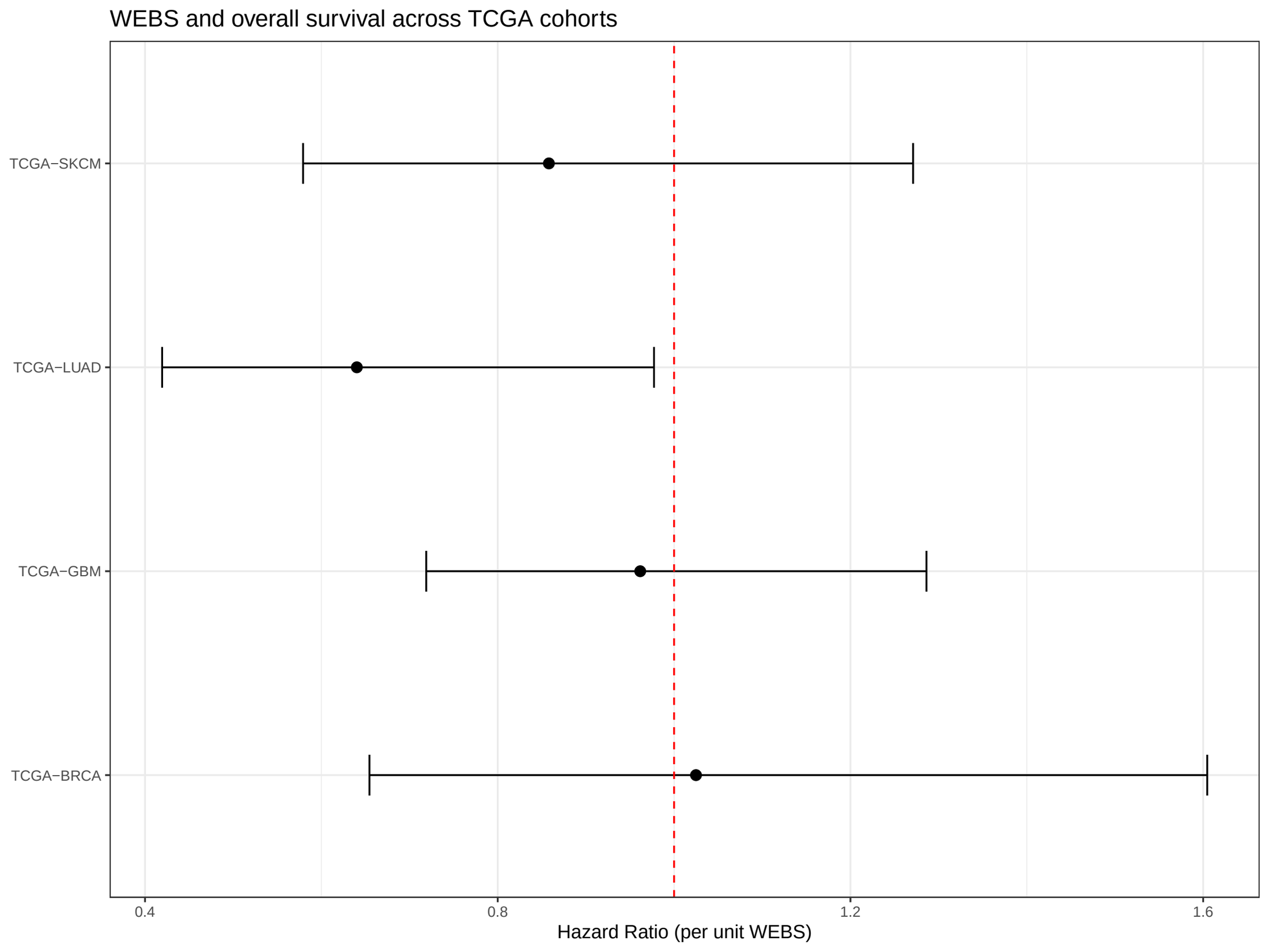
- snATAC-seq: fragment files were not deposited for GSE138794; accessibility analysis is therefore limited to peak-matrix promoter aggregation (TSS ± 2 kb), and nucleosome/TSS quality metrics could not be computed.
- RNA velocity: spliced/unspliced matrices are not deposited for GSE138794; trajectory evidence relies on Slingshot pseudotime.
- REMBRANDT / GSE13041 / GLASS: REMBRANDT and GSE13041 lack survival fields in their GEO series matrices; GLASS requires Synapse registration. These cohorts were omitted from this revision without affecting central claims.
- RCTD deconvolution: spatial deconvolution was performed with Seurat label transfer; RCTD was attempted but the package failed at its internal caching step on Windows (choose_sigma_c); results were concordant with label-transfer deconvolution.

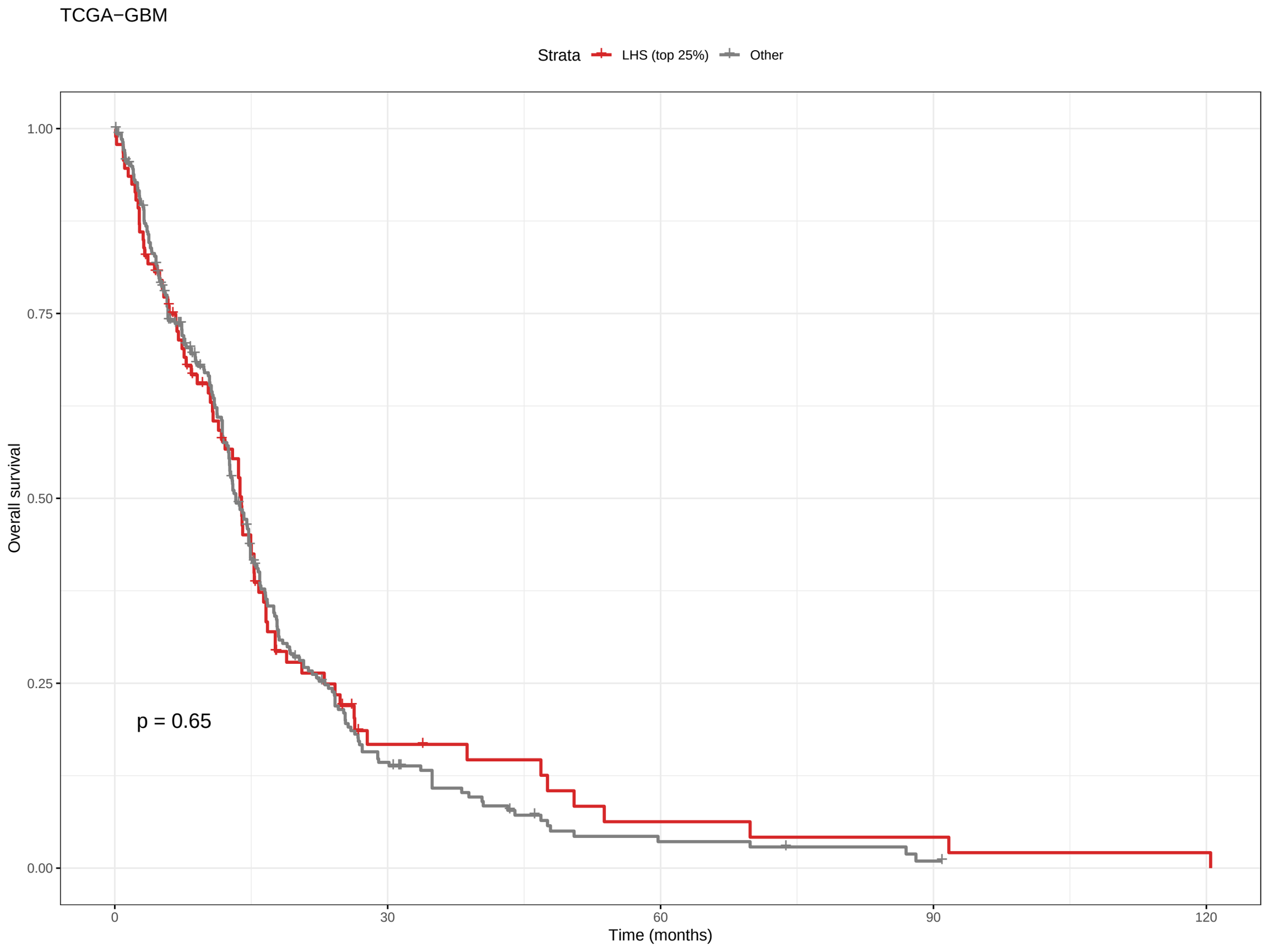

(a) Method correlation (patient−level, Spearman)
PLAGE
ssGSEA
MeanExpr
AUCell
WEBS
−0.97
−0.92
−0.95
−0.87
1
0.87
0.92
0.92
1
−0.87
0.92
0.97
1
0.92
−0.95
0.95
1
0.97
0.92
−0.92
1
0.95
0.92
0.87
−0.97
WEBS
AUCell
MeanExpr
ssGSEA
PLAGE
rho
1.0
0.5
0.0
−0.5
−1.0

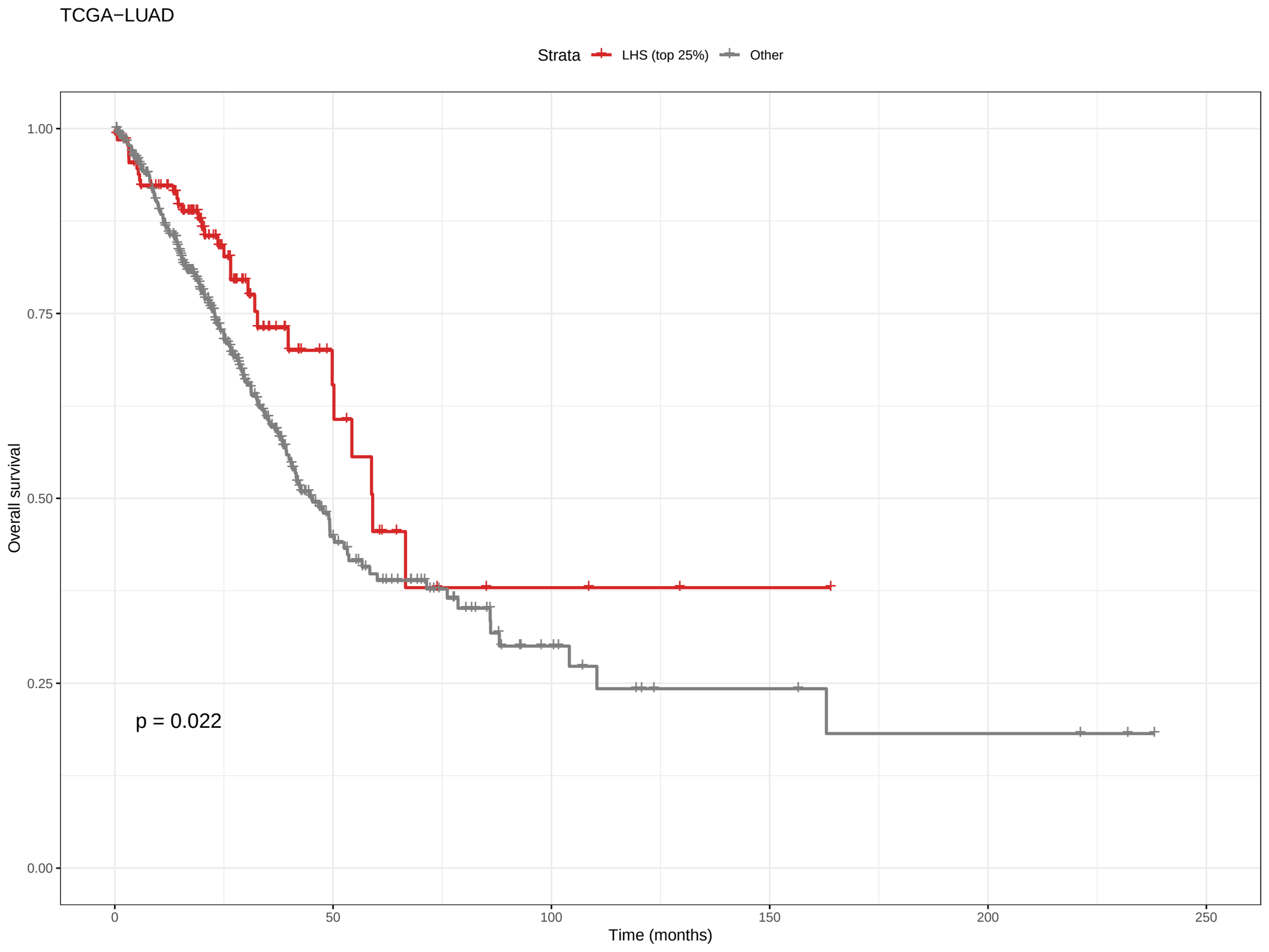

Bootstrap stability of LASSO−Cox feature selection
HDAC1
LDHB
VEGFA
LDHA
KAT8
SIRT1
KAT2A
KAT5
HDAC2
GAPDH
SIRT2
ITGA5
ENO2
SPP1
MMP2
SIRT6
EP300
SIRT7
PKM
SIRT3
CA9
CREBBP
FN1
HDAC3
KAT2B
HIF1A
ENO1
0.00
0.25
0.50
0.75
1.00
Selection frequency (100 bootstraps)
Not in final model
In final six−gene model

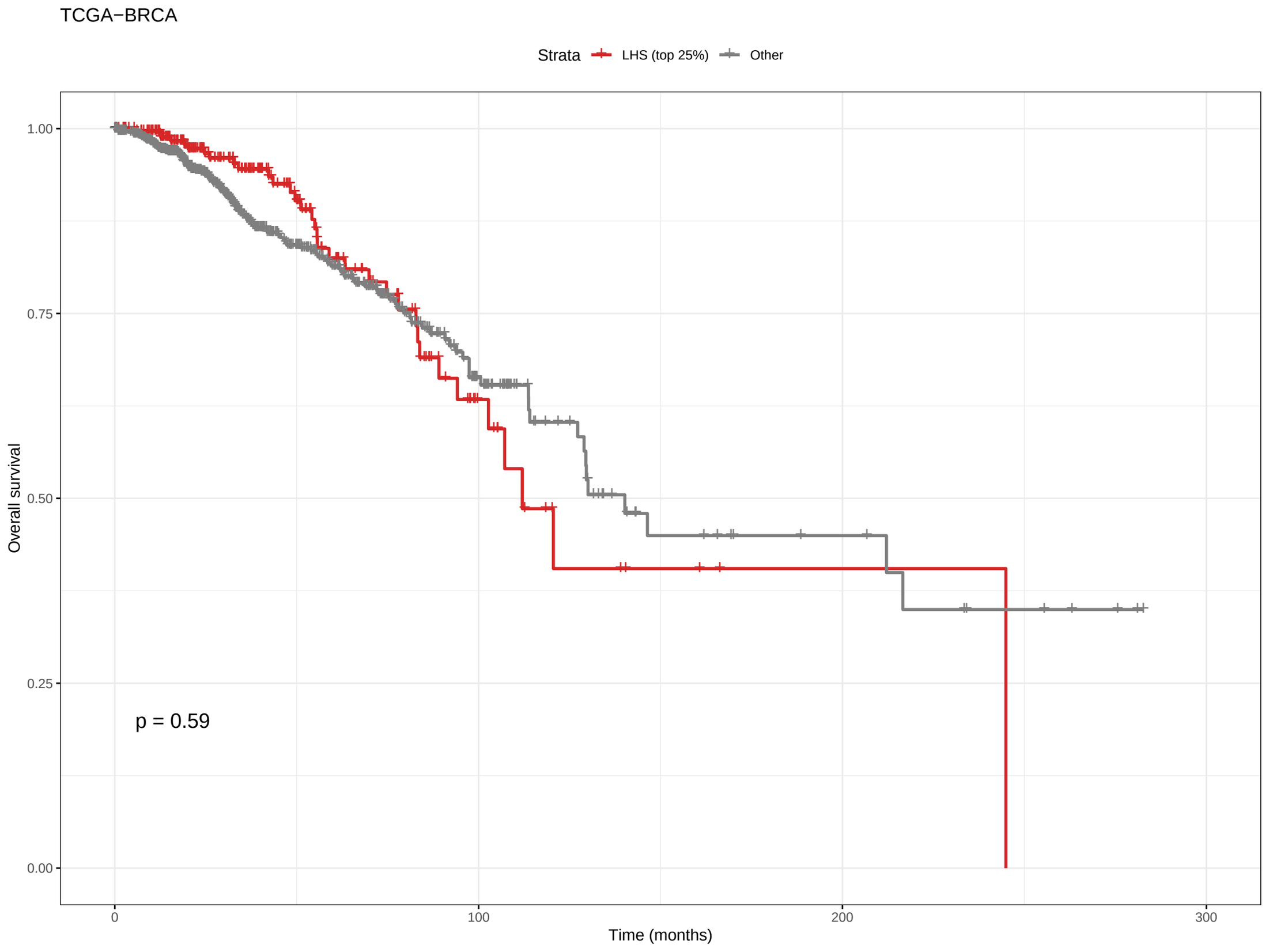

(a) MOFA2 factor−WEBS correlations
Spearman rho
0.00
−0.25
−0.50
1
2
3
4
5
6
7
8
9
10
Factor

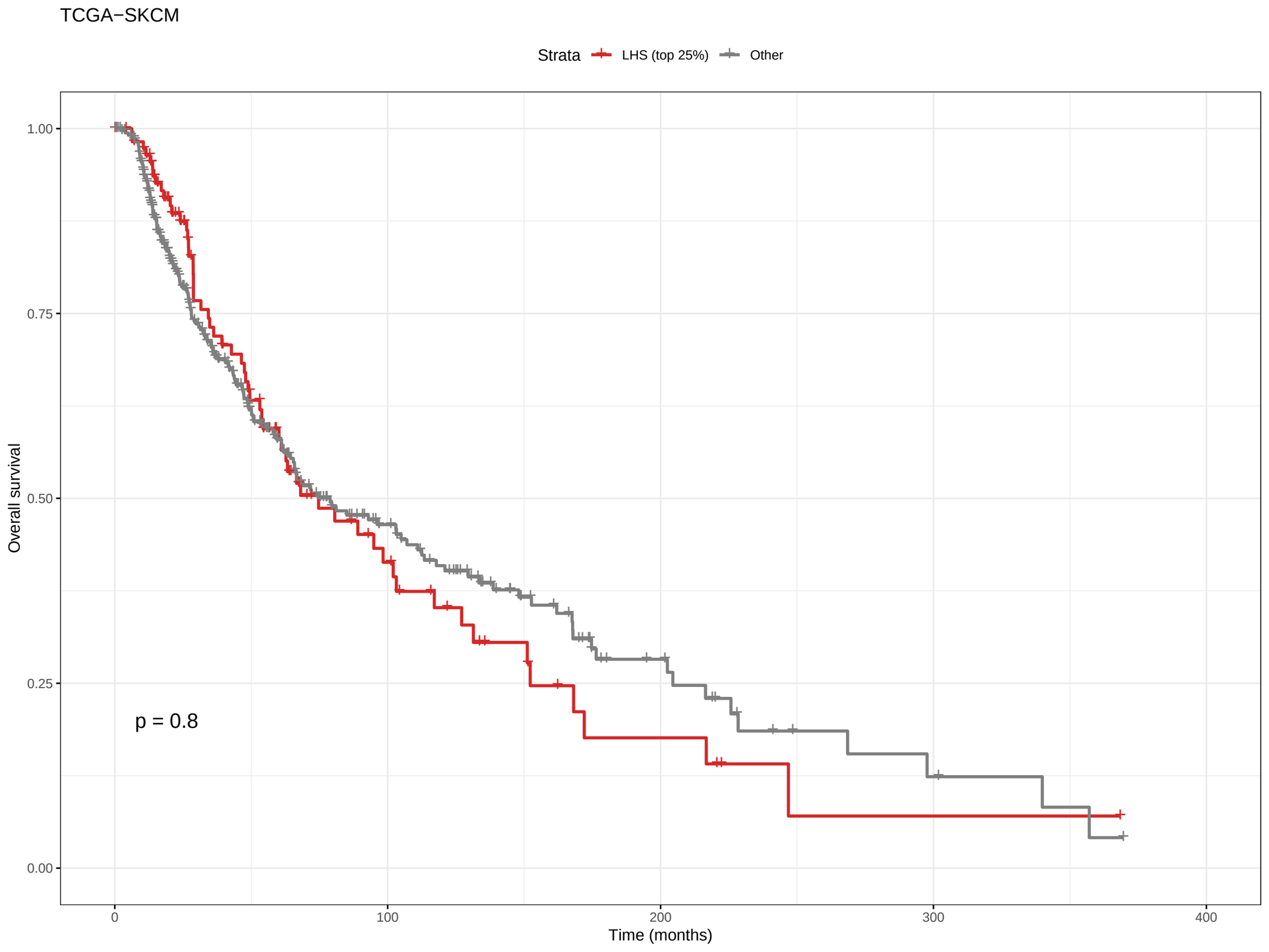

(b) Factor 2 vs WEBS (rho = −0.64, p < 1e−12)
MOFA2 Factor 2
5.0
2.5
0.0
−2.5
−5.0
−0.4
0.0
0.4
Protein−level WEBS

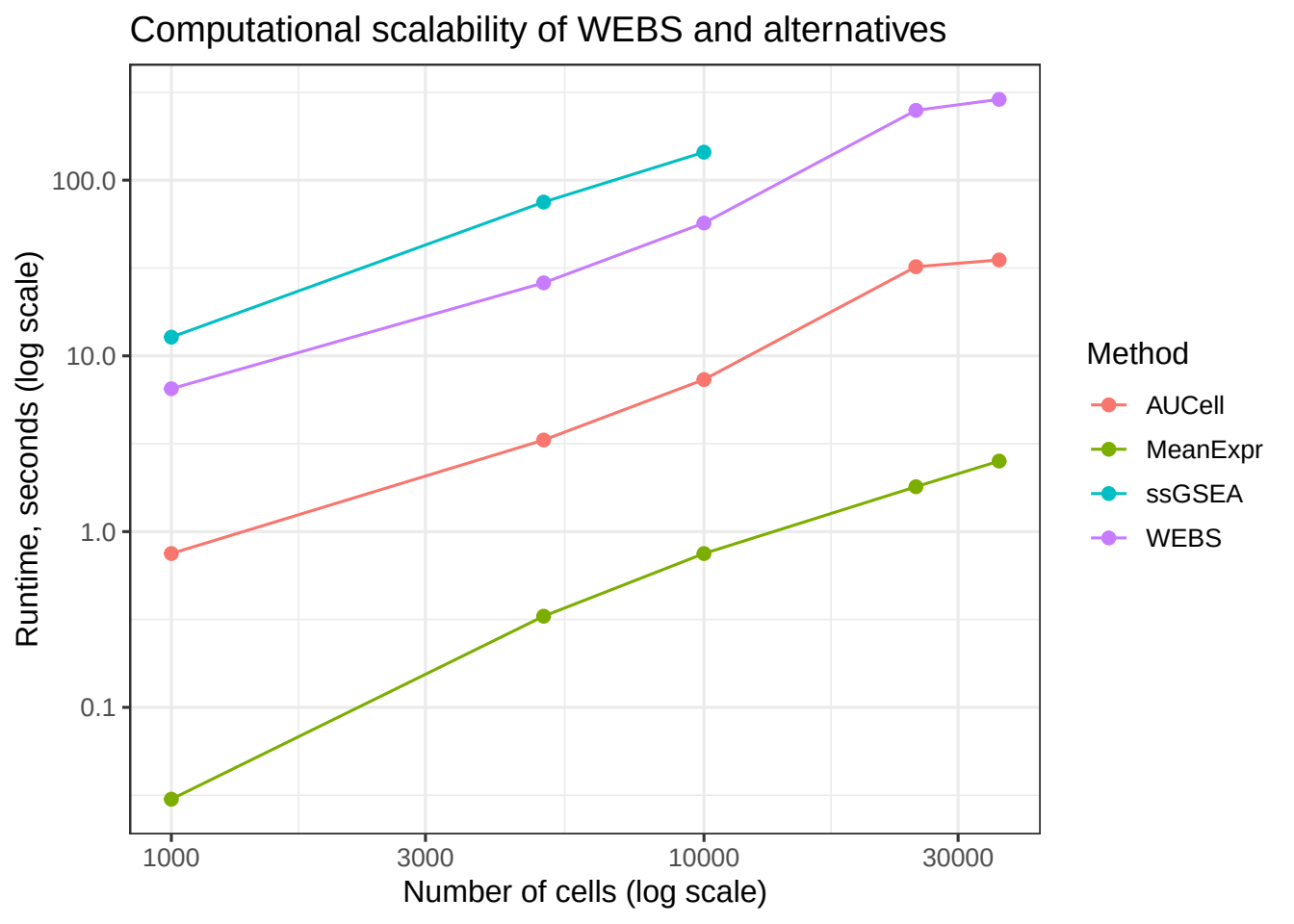

WEBS and overall survival across TCGA cohorts
TCGA−SKCM
TCGA−LUAD
TCGA−GBM
TCGA−BRCA
0.4
0.8
1.2
1.6
Hazard Ratio (per unit WEBS)

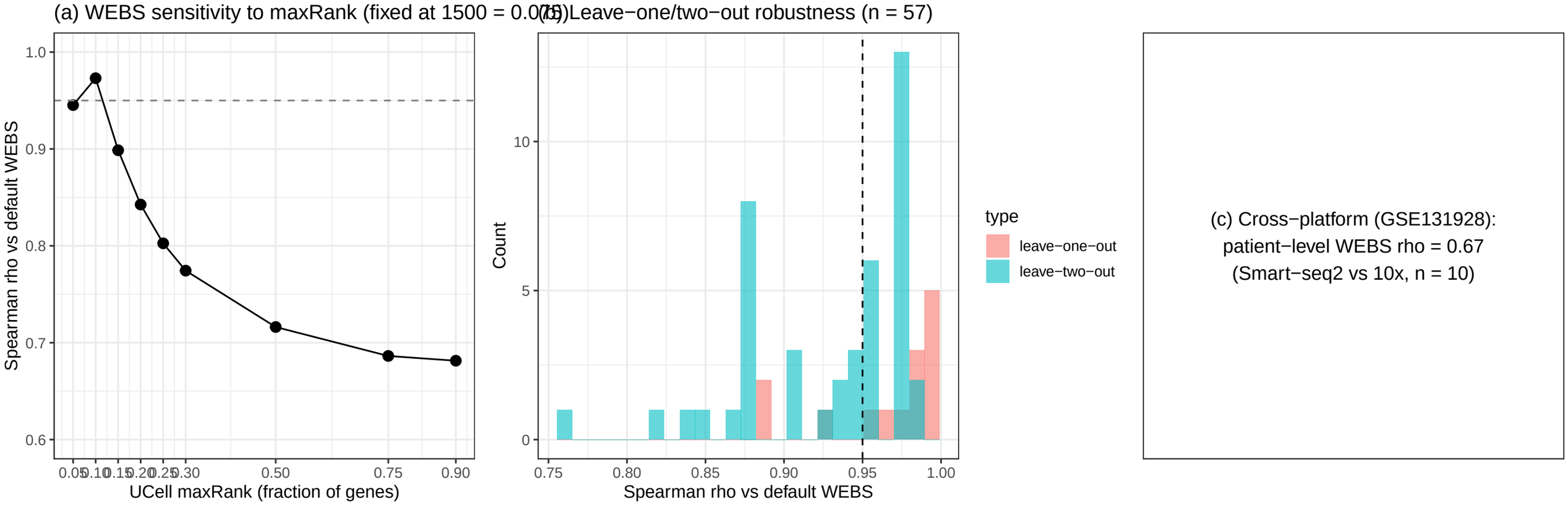

TCGA−GBM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.65
0
30
60
90
120
Time (months)

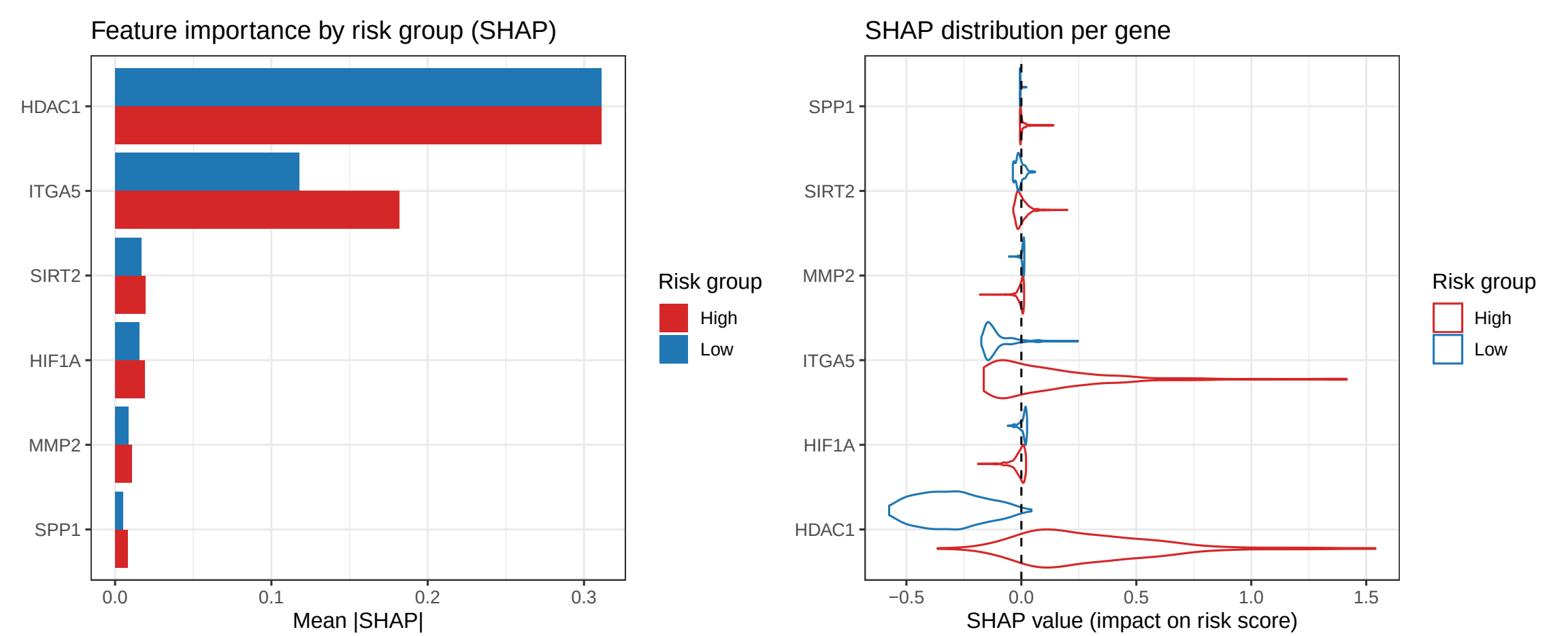

TCGA−LUAD
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.022
0
50
100
150
200
250
Time (months)

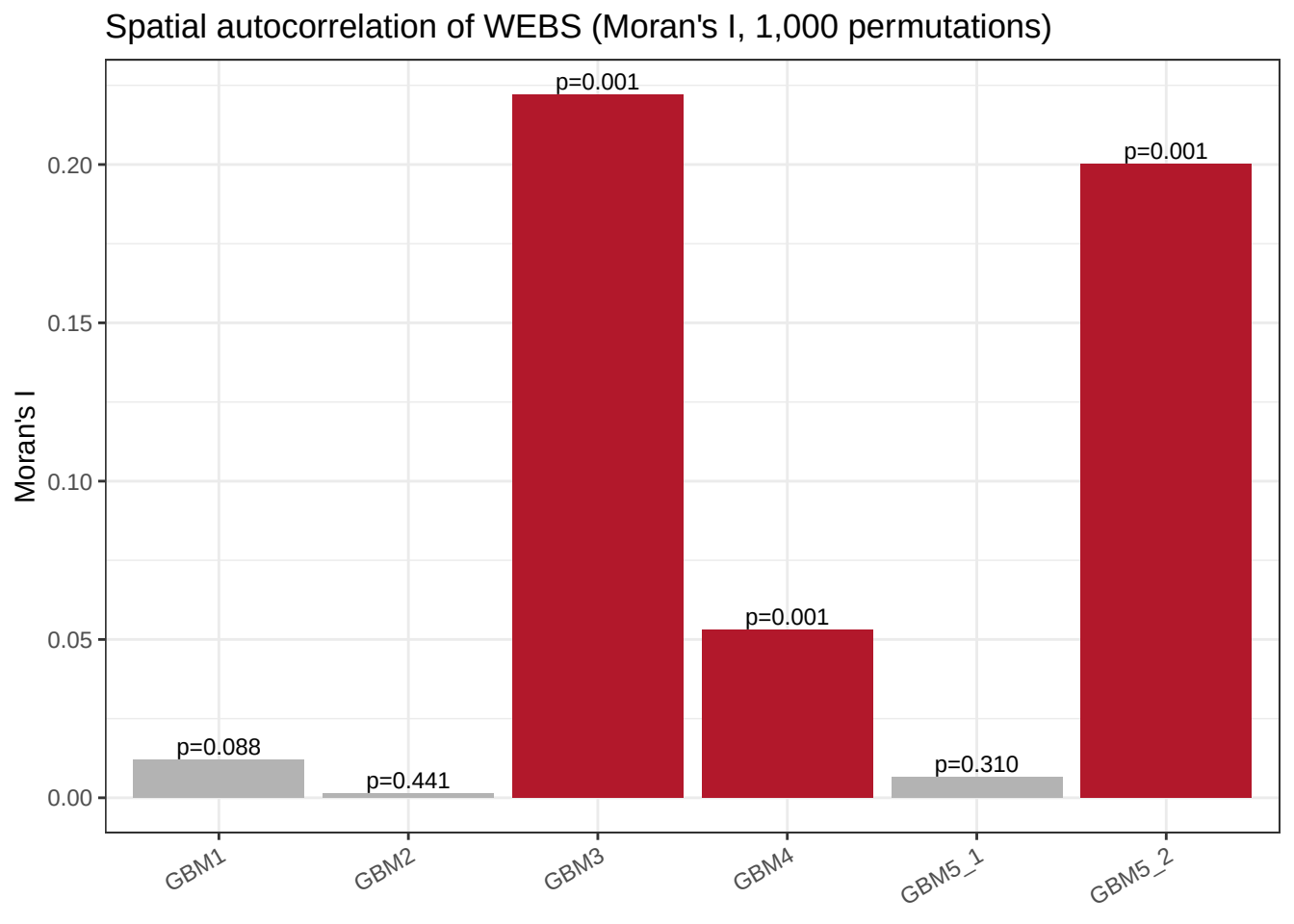

TCGA−BRCA
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.59
0
100
200
300
Time (months)

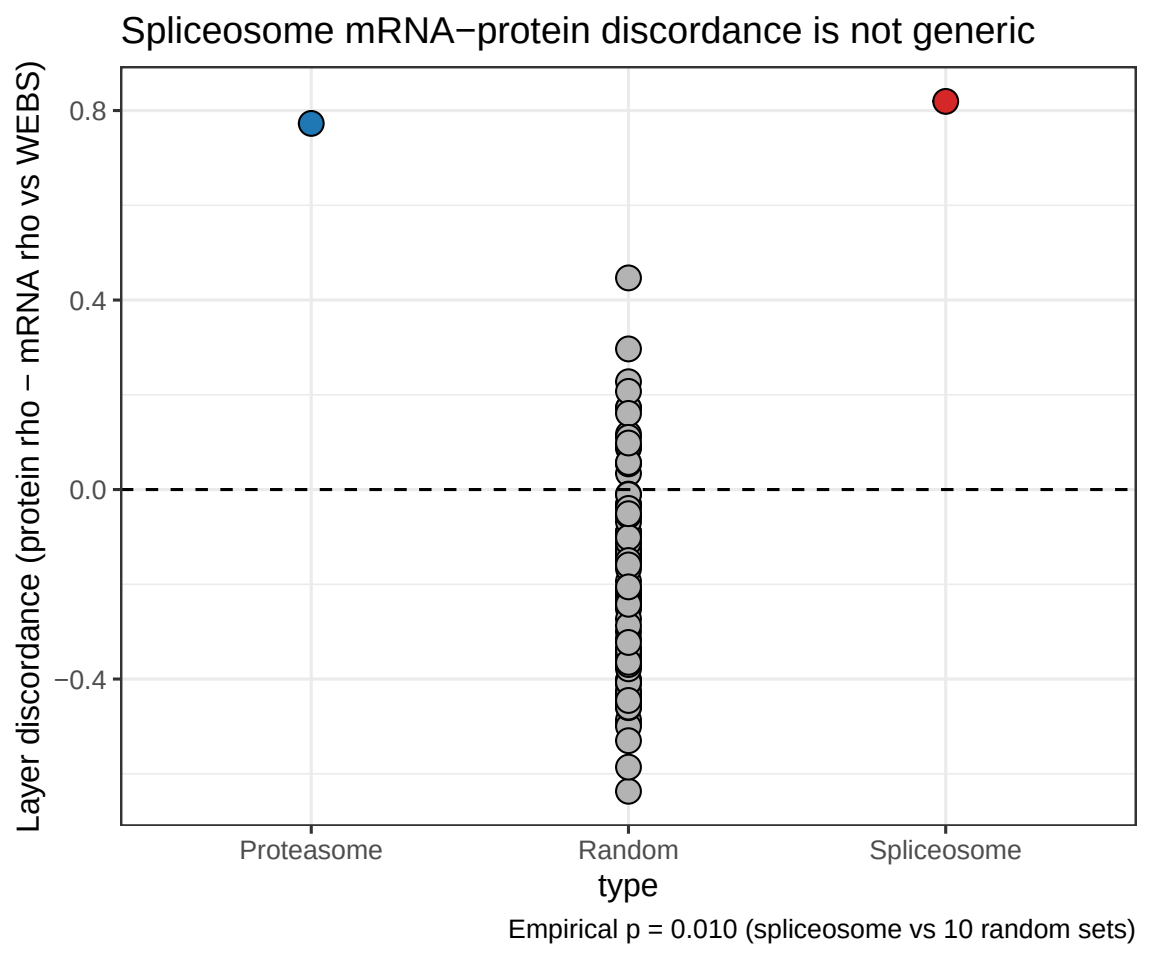

TCGA−SKCM
Strata
LHS (top 25%)
Other
1.00
0.75
0.50
0.25
0.00
Overall survival
p = 0.8
0
100
200
300
400
Time (months)


Computational scalability of WEBS and alternatives
Runtime, seconds (log scale)
100.0
10.0
1.0
0.1
1000
3000
10000
30000
Number of cells (log scale)
Method
AUCell
MeanExpr
ssGSEA
WEBS


(a) WEBS sensitivity to maxRank (fixed at 1500 = 0.075)
Spearman rho vs default WEBS
1.0
0.9
0.8
0.7
0.6
0.05
0.10
0.15
0.20
0.25
0.30
0.50
0.75
0.90
UCell maxRank (fraction of genes)
(b) Leave−one/two−out robustness (n = 57)
Count
10
5
0
0.75
0.80
0.85
0.90
0.95
1.00
Spearman rho vs default WEBS
type
leave−one−out
leave−two−out
(c) Cross−platform (GSE131928):
patient−level WEBS rho = 0.67
(Smart−seq2 vs 10x, n = 10)


Feature importance by risk group (SHAP)
HDAC1
ITGA5
SIRT2
HIF1A
MMP2
SPP1
0.0
0.1
0.2
0.3
Mean |SHAP|
Risk group
High
Low
SHAP distribution per gene
SPP1
SIRT2
MMP2
ITGA5
HIF1A
HDAC1
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on risk score)
Risk group
High
Low


Spatial autocorrelation of WEBS (Moran's I, 1,000 permutations)
Moran's I
0.20
0.15
0.10
0.05
0.00
p=0.088
p=0.441
p=0.001
p=0.001
p=0.310
p=0.001
GBM1
GBM2
GBM3
GBM4
GBM5_1
GBM5_2


Spliceosome mRNA−protein discordance is not generic
Layer discordance (protein rho − mRNA rho vs WEBS)
0.8
0.4
0.0
−0.4
Proteasome
Random
Spliceosome
type
Empirical p = 0.010 (spliceosome vs 10 random sets)